TAMPINES MERIDIAN JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

CANDIDATE NAME	( )
CIVICS GROUP	

H2 PHYSICS 9478/02

Paper 2 Structured Questions **16 September 2026**

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and Civics Group in the spaces at the top of the page.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
The use of an approved scientific calculator is expected, where appropriate.

Answer **all** questions.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiners' Use	
1	/ 7
2	/ 7
3	/ 8
4	/ 6
5	/ 9
6	/ 10
7	/ 8
8	/ 20
Deduction	
Total	/ 75

Data

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_o	=	$4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	ε_0	=	8.85×10^{-12} F m^{-1}
	$(\frac{1}{4\pi\varepsilon_0}$	=	8.99×10^{9} m F^{-1})
elementary charge	e	=	1.60×10^{-19} C
the Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
the Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
the Boltzmann constant	k	=	1.38×10^{-23} J K^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

Formulae

uniformly accelerated motion	s	=	$ut + \frac{1}{2}at^2$
	v^2	=	$u^2 + 2as$
work done on/by a gas	W	=	$p\Delta V$
hydrostatic pressure	p	=	$\frac{F}{A}$
gravitational potential	ϕ	=	$-\frac{GM}{r}$
temperature	T /K	=	T /°C + 273.15
pressure of an ideal gas	p	=	$\frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean kinetic energy of a molecule of an ideal gas	E	=	$\frac{3}{2}kT$
displacement of particle in s.h.m.	x	=	$x_o \sin \omega t$

velocity of particle in s.h.m.	v	=	$v_o \cos \omega t$
		=	$\pm \omega \sqrt{x_o^2 - x^2}$
electric current	I	=	$Anvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R}$	=	$\frac{1}{R_1} + \frac{1}{R_2} + \ldots$
capacitors in series	$\frac{1}{C}$	=	$\frac{1}{C_1} + \frac{1}{C_2} + \ldots$
capacitors in parallel	C	=	$C_1 + C_2 + \ldots$
energy in a capacitor	U	=	$\frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	Q	=	$Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
Discharging a capacitor	Q	=	$Q_0 e^{-\frac{t}{\tau}}$
RC time constant	τ	=	RC
electric potential	V	=	$\frac{Q}{4\pi\varepsilon_0 r}$
alternating current/voltage	X	=	$x_o \sin \omega t$
magnetic flux density due to a long straight wire	B	=	$\frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	B	=	$\frac{\mu_0 N I}{2r}$
magnetic flux density due to a long solenoid	B	=	$\mu_0 n I$
energy states for quantum particle in a box	E_n	=	$\frac{h^2}{8mL^2}n^2$
radioactive decay	x	=	$X_0 e^{-\lambda t}$
decay constant	λ	=	$\frac{\ln 2}{t_{\frac{1}{2}}}$

Answer **all** the questions in the spaces provided.

1 A light beam of length 1.5 m is supported by a horizontal cable placed at 0.6 m from the hinge. The beam is hinged at one end supported by the wall. A weight of 300 N is hung at the other end of the beam as shown in Fig. 1.

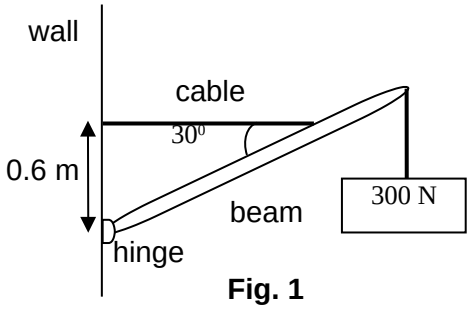


Fig. 1

(a) The three forces acting on the beam are the tension in the cable *T*, the force by the hinge on the beam *H* and the weight of 300 N.

Draw a vector diagram to represent the forces acting on the beam.

[2]

(b) Determine the magnitude of the tension in the cable *T*.

T = …… N [2]

(c) Hence determine the magnitude and direction of the force *H*.

magnitude of force *H* = …….. N

direction of force *H* : …….. [3]

2 A car of mass 1800 kg accelerates from rest on a horizontal surface.

The graph in Fig. 2.1 shows how the driving force of the car's engine F_E, and the resistive force on the car F_R, vary with the distance d travelled by the car.

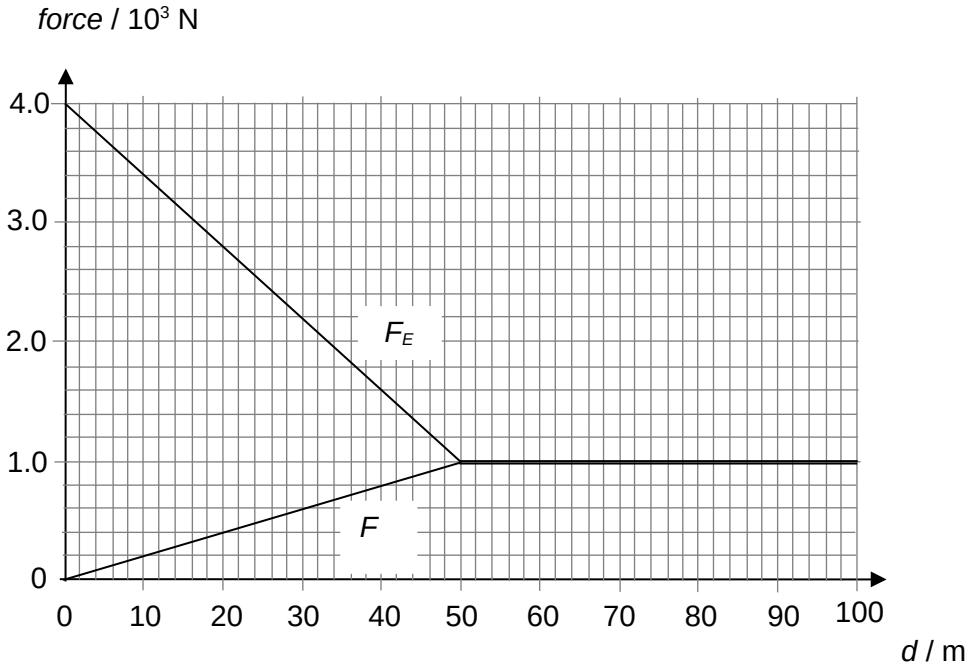


Fig. 2.1

(a) State what is represented by the area under the line F_E.

……

…… [1]

(b) **(i)** Calculate the maximum speed of the car.

maximum speed = …… m s^{-1} [3]

(ii) Calculate the power output of the engine when the car is travelling at its maximum speed.

power output = …… W [2]

(c) The mass of the car is increased when more passengers enter the car.

The power output of the motor and the resistive forces on the car remain unchanged.

Explain why the car now requires a longer time to reach the same maximum speed as that in part **(b)(i)**.

……

……. [1]

3 Fig. 3.1 shows a cannon ball fired horizontally from a cannon at the edge of a cliff. The cannon ball travels a horizontal distance of 240 m in 2.5 s before hitting the sea.

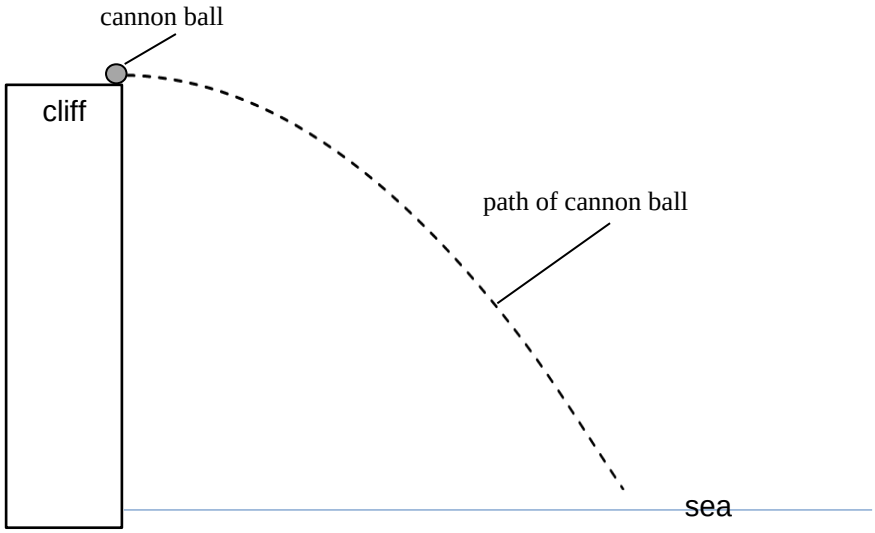


Fig. 3.1 (not to scale)

The effects of air resistance on the cannon ball are negligible.

(a) Calculate the height of cannon above sea level.

height = …….. m [2]

(b) Determine the magnitude and direction of the velocity of the cannon ball just before striking the surface of the sea.

magnitude of velocity = …… m s^{-1}

direction: ……..…… [4]

(c) The initial speed of the cannon ball is now increased. Explain the effect that this will have on the time of flight of the cannon ball.

……...

……...

……...

…… [2]

4 **(a)** State what is meant by *simple harmonic motion*.

……

……

……

…… [2]

(b) Fig. 4.1 shows a mass *m* of 0.80 kg attached to an unstretched helical spring and allowed to come to rest.

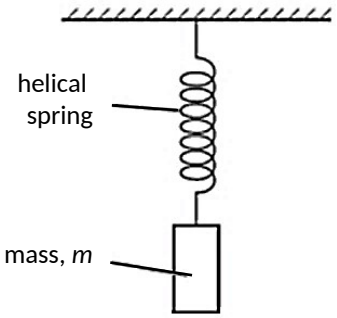


Fig. 4.1

The mass is then pulled down a distance of 1.6 cm and released, causing it to oscillate in simple harmonic motion at a frequency of 3.0 Hz.

(i) Calculate the maximum kinetic energy of the oscillations.

maximum kinetic energy = …… J [2]

(ii) Determine speed *v* of the mass at time *t* = 0.20 s.

v = …… m s^{-1} [2]

BLANK PAGE

5 **(a)** State the word equation representing the first law of thermodynamics.

……

……

……

…… [1]

(b) Fig. 5.1 shows the *P*-*V* graph of a system of ideal gas undergoing three different processes.

Along the path **ABC** the work done by the gas is 40 kJ and the heat energy supplied to the gas is 90 kJ.

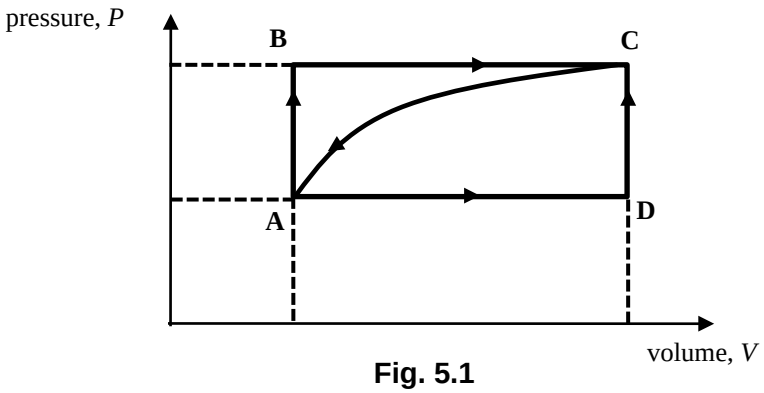


Fig. 5.1

(i) Calculate the internal energy at **C** if the internal energy at **A** is 45 kJ.

internal energy = ……. kJ [2]

(ii) Along the path **ADC** the work done by the gas is 20 kJ.

Calculate the heat energy supplied to the gas along the path **ADC**.

energy supplied = ……. kJ [2]

(iii) The work done on the gas when it returns along the curved path **CA** is 31 kJ.

1. Calculate the net work done on the gas in the cycle **ABCA**.

net work done on gas = …… kJ [2]

2. Calculate the net heat supplied to the gas in the cycle **ABCA**.

Explain your answer clearly.

……

……

net heat supplied = …… kJ [2]

6 **(a)** Define *magnetic flux*.

……
....

……
…. [1]

(b) A circular coil of 300 turns and of radius 4.0 cm is placed in a uniform magnetic field of flux density 5.0×10^{-2} T. The coil is rotated in the magnetic field and the variation with time of magnetic flux linkage ϕ of the coil is sinusoidal as shown in Fig. 6.1.

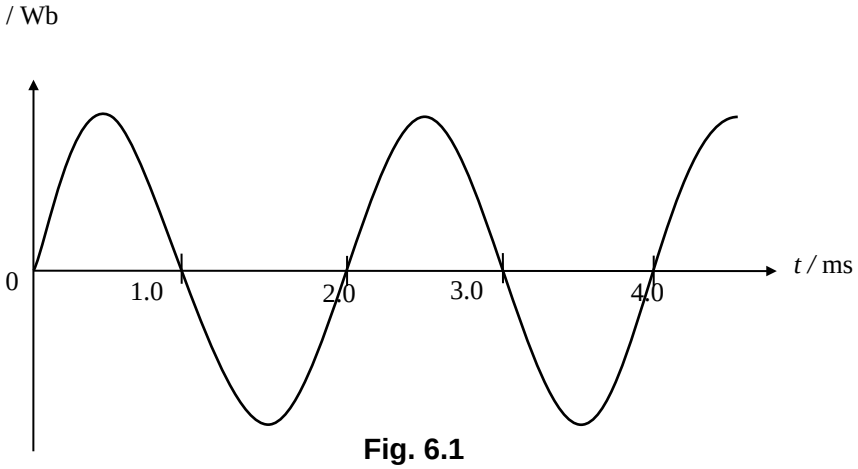


Fig. 6.1

(i) State the angle between the normal of the plane of the coil and the magnetic field at time = 0 s as shown in Fig. 6.1.

angle = …….. ° [1]

(ii) Calculate the maximum magnetic flux linkage of the coil.

maximum magnetic flux linkage = …….. Wb [2]

(iii) Calculate the average electromotive force (e.m.f.) induced across the coil when it rotates through the first quarter of a revolution.

average e.m.f. = ... V [2]

(iv) Explain how the minimum e.m.f. can be determined from Fig. 6.1.

State the first instant at which minimum e.m.f. is induced across the coil.

.....

.....

first instant of minimum e.m.f. = ... ms [2]

(v) Explain how the e.m.f. induced will vary as the coil rotates.

.....

.....

.....

..... [2]

7 The setup in Fig. 7.1 is used to investigate the photoelectric effect.

Monochromatic electromagnetic radiation of wavelength λ is incident on a metal emitter plate E. The emitted photoelectrons from E reach the collector plate C and are detected by the ammeter as a current. The potential difference across E and C can be adjusted via the variable voltage source.

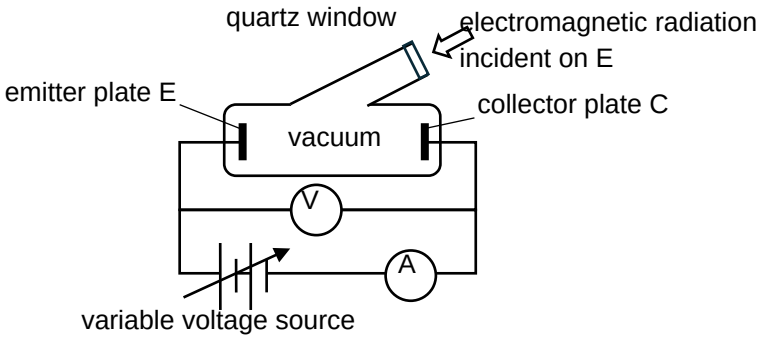


Fig. 7.1

The wavelength of the light λ is varied and the minimum potential difference V_s across E and C that results in no current being detected in the ammeter is recorded. The graph in Fig. 7.2 shows the variation of V_s with λ.

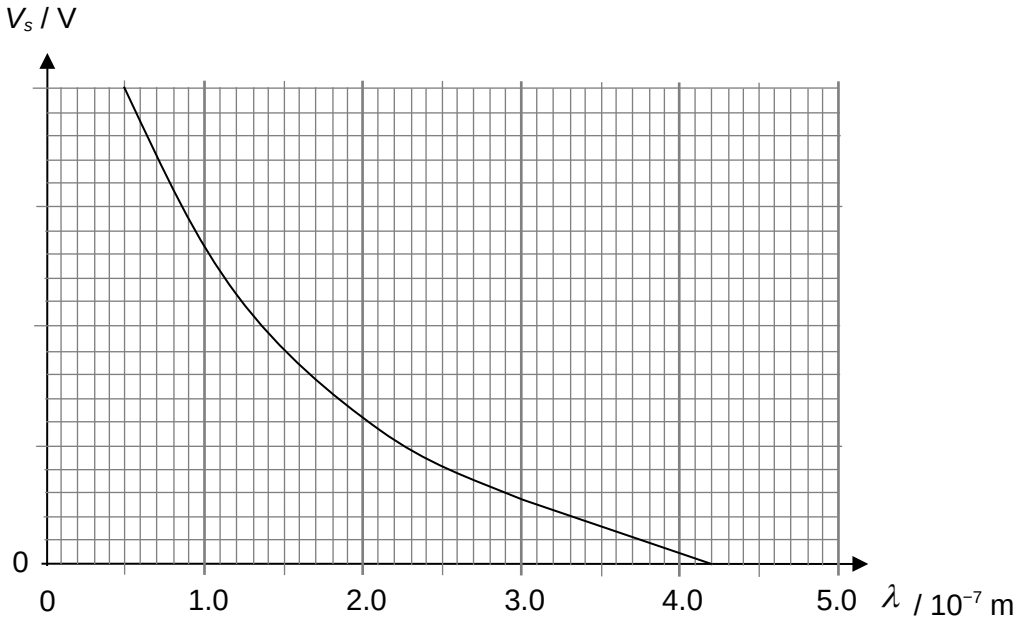


Fig. 7.2

(a) State the significance of the horizontal intercept of the graph.

……

…… [1]

(b) **(i)** Calculate the minimum photon energy required to allow an electron to overcome the surface energy barrier and be emitted from the emitter plate.

energy = …… J [2]

(ii) Determine the maximum kinetic energy of the emitted electron when the wavelength used is 2.0×10^{-7} m.

kinetic energy = …… J [2]

(c) Explain how the existence of a threshold frequency in the photoelectric effect provides evidence for the particulate nature of electromagnetic radiation.

……

……

……

……

……

……

…… [3]

8 **Semiconductors and Light-Emitting Diodes (LEDs)**

The following article describes how semiconductors are made into devices such as diodes, light emitting diodes and quantum-well light emitting diodes.

Doped Silicon Semiconductors

Some semiconductors are made of silicon which is an electrical insulator. To allow silicon to conduct electricity, a process known as doping adds impurity atoms into the silicon crystal lattice to increase the number density of charge carriers. Impurity atoms such as phosphorus create an **n-type semiconductor** where **free electrons** are the majority charge carriers. On the other hand, doping with atoms such as boron creates a **p-type semiconductor** where the majority charge carriers are **positive holes**.

A sample of an n-type silicon semiconductor is shown in Fig. 8.1. together with its dimensions.

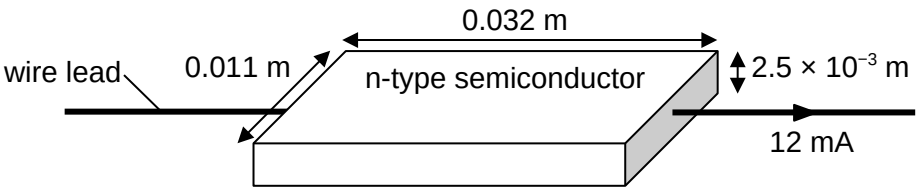


Fig. 8.1

Pure silicon has a density of 2.33×10^3 kg m^{-3} and a molar mass of 28.1×10^{-3} kg mol^{-1}.

The n-type silicon slab in Fig. 8.1 was doped with **one phosphorus atom for every 2.0×10^6 silicon atoms.** Each phosphorus atom donates one free conduction electron. In the sample above, the doping in the n-type silicon slab allows it to carry a current of 12 mA.

The p-n Junction Diode

Something interesting happens when you dope half of a silicon slab with n-type atoms and the other half with p-type atoms: you form a **p-n junction diode**.

Fig. 8.2 shows a diagram of a p-n junction diode.

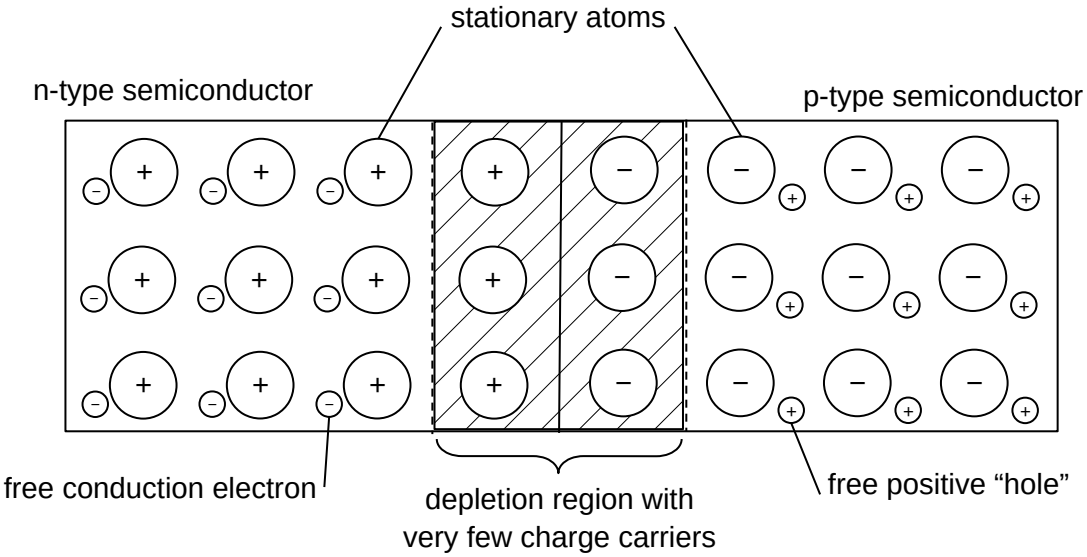


Fig. 8.2

In the central shaded region, known as the **depletion region,** the mobile electrons and holes recombine and neutralise one another, so that the shaded region has very few mobile charge carriers. A potential gradient is set up in the region, which creates an electric field that opposes the diffusion of charge carriers. The potential gradient in the depletion region of width 0.33 μm is shown in Fig. 8.3.

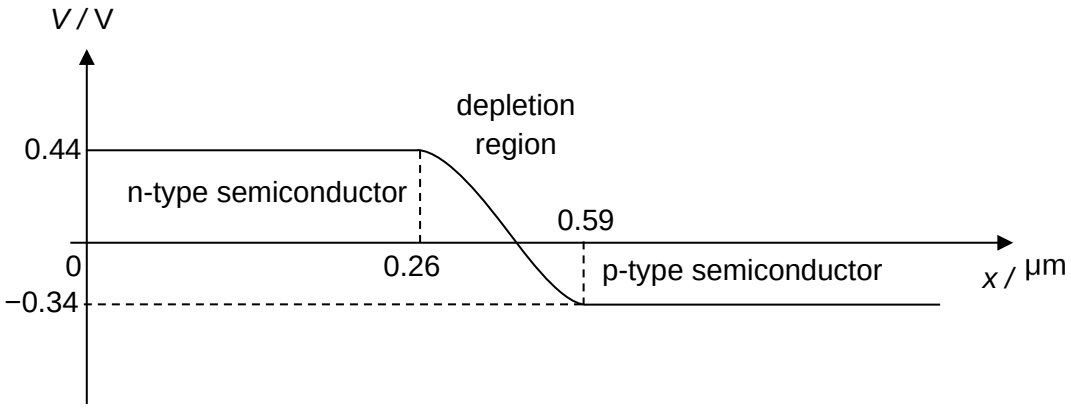


Fig. 8.3

Light-Emitting Diodes (LEDs)

Some semiconductor diodes such as Gallium Arsenide (GaAs) were found to produce light when conducting current in forward-bias mode. In these light-emitting diodes (LEDs), when electrons and holes recombine, a higher energy level electron in the conduction band will transition to a lower energy level hole in the valence band and release energy in the form of a photon.

An example of the energy levels in a GaAs LED is shown in Fig. 8.4.

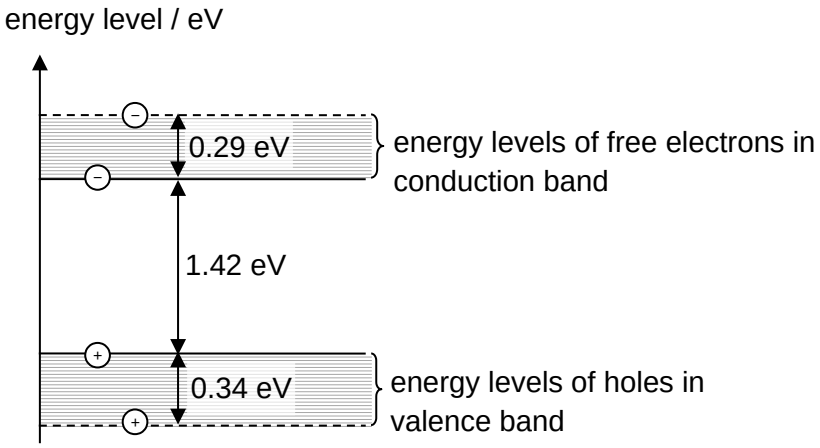


Fig. 8.4

The free electrons in the conduction band occupy a range of energy levels, as do the holes in the valence band. As a result, light emitted from LEDs is not monochromatic and is not a “pure” colour.

Quantum-Well Light Emitting Diodes (QLEDs)

The broad emission spectrum of conventional LEDs makes them unsuitable for use in devices requiring pure colors, such as in high-definition display screens. To address this shortcoming, modern display technologies employ **Quantum-Well Light-Emitting Diodes (QLEDs)**.

A quantum well is made by sandwiching a very thin layer of a semiconductor between thicker layers of another type of semiconductor. When the width of this central layer is reduced to near the de Broglie wavelength of an electron (around a few nanometers thick), quantum effects in the system become significant.

An electron trapped inside this ultra-thin semiconductor layer can be modeled as a particle trapped within a one-dimensional potential energy well, bounded by infinite potential walls. This strict confinement forces the electron's energy to be restricted to a few discrete values. A monitor screen made from such QLED pixels can thus show pictures with exceptionally pure colours.

(a) Show that the number density of free electrons in the n-type semiconductor in Fig. 8.1 is 2.5×10^{22} m^{-3}.

[2]

(b) Calculate the drift velocity of the electrons in the n-type semiconductor in Fig. 8.1.

drift velocity = ... m s^{-1} [2]

(c) With reference to the electric potential graph in Fig. 8.3, explain the direction of the electric field in the depletion region.

.....

.....

.....

..... [2]

(d) Using Fig. 8.3, calculate the average electric field strength within the depletion region.

electric field strength = ... V m^{-1} [2]

(e) The depletion region has positive and negative charges separated on either side, similar to a parallel plate capacitor. The stored charge in the depletion region is equal on both sides and has a magnitude of 3.6×10^{-8} C.

Estimate the capacitance of the depletion layer.

capacitance = ... F [2]

(f) With reference to Fig. 8.4, determine the minimum wavelength of the photons that could be produced by electron-hole recombination in the GaAs LED.

minimum wavelength = ... nm [2]

(g) State the region of the electromagnetic spectrum in which the wavelength of your answer to **(f)** lies.

……

…… [1]

(h) Fig. 8.5 below shows the energy levels in the conduction band of a quantum well semiconductor LED. The quantum well has a thickness d of 5.0×10^{-9} m and a potential energy barrier on either side of the quantum well of $E_b = 0.14$ eV.

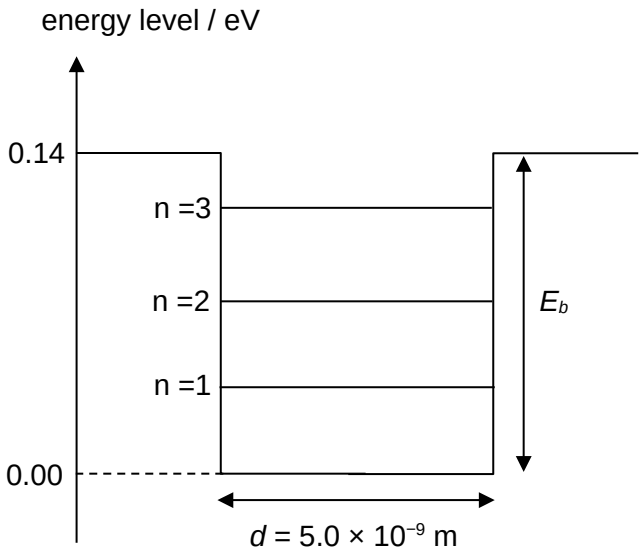


Fig. 8.5

(i) On Fig. 8.5, sketch the wavefunctions that represent the electron in the $n = 1$ and $n = 3$ energy levels.

[2]

(ii) Given that the allowed values of momentum for the electron in the nth energy level of the quantum well is given by

$$p_n = \frac{nh}{2d}$$

show that the allowed values of the energy E_n of the electron in each energy level n is given by the expression

$$E_n = \frac{n^2h^2}{8md^2}$$

[2]

(iii) Determine the energy of the $n = 3$ energy level.

Hence, explain why this is the highest discrete energy level that can be found within the quantum well.

……

……

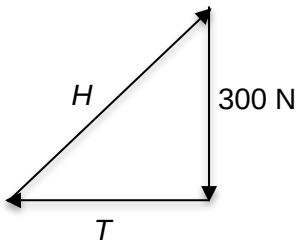
…… [3]

BLANK PAGE

Tampines Meridian Junior College

2026 JC2 H2 Physics Preliminary Examination Paper 2

Suggested Solution

1	**(a)**	 [M1] for correct vector triangle with correct direction [A1] for correct labels of forces [deduct one mark for label angle 30°]
	(b)	By taking moments about the hinge, $T \times 0.60 = 300 \times 1.5 \cos 30^\circ$ [C1] $T = 650$ N [A1]
	(c)	Horizontally, $\Sigma F_x = 0$ $T = H_x$, $H_x = 650$ N Vertically, $\Sigma F_y = 0$ $H_y = 300$ N [C1 for both equations] $H = \sqrt{H_x^2 + H_y^2} = \sqrt{650^2 + 300^2}$ $= 716$ N [A1] $\theta = \tan^{-1}\left(\frac{H_y}{H_x}\right)$ $= \tan^{-1}\left(\frac{300}{650}\right)$ $= 24.8^\circ$ [A1] 24.8° anticlockwise above the horizontal.

2	**(a)**	The area under F_E represents the work done by the driving force of the car's engine. [A1]	
	(b)	**(i)**	Net work done = area bounded by F_E and F_R $= \frac{1}{2}(50)(4000) = 1.0 \times 10^5$ J [C1] This is also the gain in the kinetic energy of the car. Since the car started from rest, the maximum speed of the car is: $\frac{1}{2}(1800)v^2 = 100000$ [M1] $v = 10.5$ m s^{-1} [A1]
		(ii)	After d = 50 m, the car is at maximum speed. The power of the car at that point is $P = Fv = (1000)(10.54)$ [M1] $= 1100$ W [A1] Ecf allowed
	(c)		Since the maximum speed is unchanged but the mass has increased, the final kinetic energy of the car is increased. Since the power of the engine is the same, more time is now needed to for the car to gain this larger amount of kinetic energy.

3	**(a)**	Vertical component of the initial velocity, $u_y = 0$ $s_y = \frac{1}{2}a_y t^2$ $= \frac{1}{2}(9.81)(2.5)^2$ $= 31 \text{ m}$	[M1] [M1] [A0]
	(b)	$u_x = v_x = \frac{240}{2.5} = 96.0 \text{ m s}^{-1}$ $v_y = a_y t = (9.81)(2.5) = 24.5 \text{ m s}^{-1}$ Resultant velocity, $v = \sqrt{(96.0)^2 + (24.5)^2} = 99.1 \text{ m s}^{-1}$ Direction, $\theta = \tan^{-1}\frac{24.5}{96.0} = 14.3^\circ$ clockwise below horizontal	[C1] [C1] [A1] [A1]
	(c)	Even though the initial speed had increased, the initial vertical component of the speed remains the same. Hence there is no change to the time of flight.	[M1] [A1]

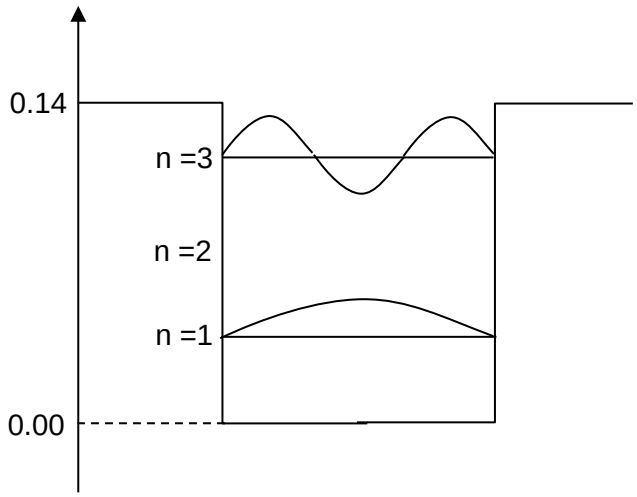
4	**(a)**	Simple harmonic motion (SHM) is defined as the oscillation of a particle such that its <u>acceleration is proportional to its displacement</u> from the equilibrium position, and is always <u>directed towards the equilibrium position</u> (or opposite in direction to the displacement).		B1 B1
	(b)	**(i)**	$\omega = 2\pi f = 2\pi(3.0) = 18.85 \text{ rads}^{-1}$ Maximum kinetic energy = ½ $m\omega^2 x_o^2$ = ½ (0.80)(18.85) ^{2}(0.016)^2 = 0.0364 J ≈ 0.036 J	C1 A1
		(ii)	$x = -x_0 \cos \omega t$ $v = \frac{dx}{dt} = \omega x_0 \sin \omega t$ $\omega = 2\pi f = 2\pi(3.0) = 18.85 \text{ rads}^{-1}$ Velocity v = (18.85)(0.016) sin(18.85(0.20)) Velocity v = -0.1773 m s^{-1} Speed v ≈ 0.18 m s^{-1} (Do not accept negative sign for final answer.)	C1 A1

5	**(a)**	Increase in internal energy = work done on the system + heat supplied to the system. One mark for correct definition [M1] One mark for putting into a "word equation" [A1]			
	(b)	**(i)**	$u_C = u_A + \Delta u$ $= 45 + (-40 + 90)$ $= 95$ kJ		[C1] [A1]
		(ii)	$\Delta u = Q + W$ $-40 + 90 = Q - 20$ $Q = 70$ kJ		[C1] [A1]
		(iii)	**1.**	net work done on gas = -40 + 31 = -9.0 kJ	[C1] [A1]
		(iii)	**2**	Since there is no change in internal energy in a cycle, $\Delta u = 0$ $Q_{net} + W_{net} = 0$ $Q_{net} - 9.0 = 0$ $Q_{net} = 9.0$ kJ Ecf allowed.	[A1] [A1]

6	**(a)**	Magnetic flux is defined as the product of magnetic flux density and the cross-sectional area perpendicular to the direction of the magnetic flux density. (from LO) Or Magnetic flux is the product of the flux density normal to the surface and the area of the surface. (from notes). [1]	
	(b)	**(i)**	90° or 270° [1]
		(ii)	Max Φ = NBA $= 300 \times 5.0 \times 10^{-2} \times \pi(4.0 \times 10^{-2})^2$ [C1] = 0.0754 Wb [A1]
		(iii)	Average emf $= \frac{\Delta \Phi}{\Delta t}$ $= \frac{(0.0754-0)}{0.50 \times 10^{-3}}$ [C1] $= 151\,\text{V}$ [A1]
		(iv)	The minimum e.m.f. can be determined from the time at which the gradient of the graph is zero. [1] The first instant is at t = 0.50 ms. (no marks deducted for sf) [1]
		(v)	As the coil rotates, the magnetic flux linkage of the coil varies sinusoidally. By Faraday's Law, the induced e.m.f is directly proportional to the rate of change of magnetic flux linkage. [M1] Hence the e.m.f will also vary sinusoidally. [A1]

7	**(a)**	The horizontal intercept of the graph shows the <u>threshold wavelength</u> above which no photoemission of electrons will take place. [A1] Accept <u>threshold frequency</u> if a reference is made to $v = f\lambda$ Also accept: <u>Maximum wavelength</u> above which <u>no emission</u> of electrons will take place.	
	(b)	**(i)**	The minimum energy required can be found when the photon with the threshold wavelength causes photoemission – this is the x-axis intercept of the graph. Photon energy at threshold wavelength $= \frac{hc}{\lambda}$ $= \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{4.2 \times 10^{-7}}$ [C1] $= 4.74 \times 10^{-19}$ J $= 4.7 \times 10^{-19}$ J [A1]
		(ii)	The kinetic energy of the emitted electron will be: $KE = \frac{hc}{\lambda} - \phi$ $= \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{2.0 \times 10^{-7}} - 4.74 \times 10^{-19}$ [C1] $= 5.21 \times 10^{-19}$ J [A1]
	(c)	Light is made of discrete <u>quanta of electromagnetic radiation</u> / <u>packets of energy</u> called <u>photons.</u> [B1] <u>One electron</u> can only <u>absorb</u> the energy of <u>one photon</u> at a time. [B1] Photons with wavelengths <u>longer than the threshold wavelength</u> / <u>frequency lower than the threshold frequency</u> <u>do not have sufficient energy</u> [B1] for electrons to overcome the surface energy barrier to escape the metal.	

8	**(a)**	Number of silicon atoms per m^3: $n_{silicon} = \frac{2.33 \times 10^3}{0.0281} \times 6.02 \times 10^{23} = 4.99 \times 10^{28}$ [C1] Number of phosphorus atoms per m^3: $n_{phosphorus} = \frac{4.99 \times 10^{28}}{2 \times 10^6}$ [M1] $= 2.50 \times 10^{22}$ [A0]
	(b)	Using $I = nAvq$ $0.012 = 2.5 \times 10^{22} \times 0.25 \times 10^{-2} \times 1.1 \times 10^{-2} \times v \times 1.6 \times 10^{-19}$ [M1] $v = 0.109 \text{ m s}^{-1}$ [A1]
	(c)	The electric field strength is the negative of the potential gradient. Since the potential is <u>decreasing towards the right</u>, or <u>Fig. 8.3 showed negative gradient (together with electric field strength is the negative of the potential gradient)</u>, [M1] the electric field is pointing <u>left to right</u>. [A1]
	(d)	$E = \left\lvert \frac{\Delta V}{\Delta x} \right\rvert$ $= \frac{0.44 - (-0.34)}{(0.59 - 0.26) \times 10^{-6}}$ [M1] $= 2.36 \times 10^6 \text{ V m}^{-1}$ [A1]
	(e)	$C = \frac{Q}{V}$ $= \frac{3.6 \times 10^{-8}}{0.44 - (-0.34)}$ [M1] $= 4.62 \times 10^{-8} \text{ F}$ [A1]
	(f)	minimum wavelength: $\lambda = \frac{hc}{(0.29 + 1.42 + 0.34)e}$ [M1] $= 606 \times 10^{-9} \text{ m}$ [A1]
	(g)	<u>red/orange light</u> for 606 nm [B1] <u>infrared</u> for 875 nm [allow ecf from part (e)]

	(h)	**(i)**	energy level / eV  [B1] for n = 1 [B1] for n = 3
		(ii)	kinetic energy $E = \frac{p^2}{2m}$ [C1] $E = \frac{\left(\frac{nh}{2d}\right)^2}{2m}$ [M1] $E = \frac{n^2h^2}{8md^2}$ [A0]
		(iii)	When n = 3 $E = \frac{(3)^2\left(6.67\times10^{-34}\right)^2}{8\times9.11\times10^{-31}\times\left(5\times10^{-9}\right)^2}$ [M1] $=2.17\times10^{-20}$ $=0.136$ eV [A1] This is very close to the potential height of the quantum well of 0.14 eV. The <u>next energy level</u> is greater, almost double due to the coefficient of 4^2 and so <u>will exceed the energy barrier</u>, and the electron with that amount of energy will not be confined in the well. [B1]